Michael Galileo Hasian (201344020)	**Rangkuman Minggu ke-3**	Senin, 9 September 2022
		Aplikasi Mikrokontroler
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Program Konversi Desima ←→ Oktal

1. Buat program Konversi Data Desimal 2 digit (yang ada di R20) ke Oktal. Simpan hasil konversi pada R22:R21.

- **Kode Program (Metode Penyisipan 0)**

```
.include "m8535def.inc"
.org 0x00
rjmp main
      ; PAKAI PROJECT "ROOT"
main :
ldi r20,low(ramend)
out spl,r20
ldi r20,high(ramend)
out sph,r20

ldi r20,0x69                ; register input data
ldi r18,0x0a                ; faktor pengali 0A
mov r16,r20
andi r16,0xf0               ; mengambil nilai puluhan
lsr r16
lsr r16
lsr r16
lsr r16
mul r16,r18                         ; puluhan kali 0A
mov r16,r20
andi r16,0x0f
add r0,r16
```

```
mov r17,r0
andi r17,0x07  ;mengambil 3 LSB terkahir; awal proses penyisipan 0 pertama
mov r21,r17
mov r17,r0
andi r17,0xf8                ; mengambil 5 MSB awal
mov r25,r17
add r25,r25
add r25,r21                          ; akhir proses penyisipan 0 pertama

mov r17,r25
andi r17,0x80                ; mengambil 1 MSB
mov r21,r17
mov r17,r25
andi r17,0x7f                ; mengambil 7 Bit terkahir
mov r23,r17

add r21,r21
adc r22,r31
add r21,r23

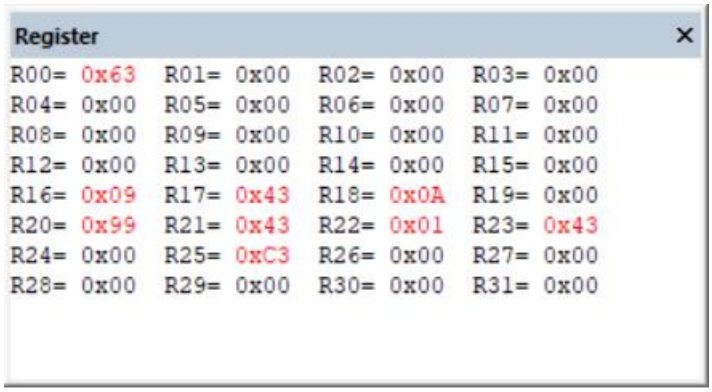
akhir: rjmp akhir
```

- **Output Program**



Register Input Data : R20 = 69

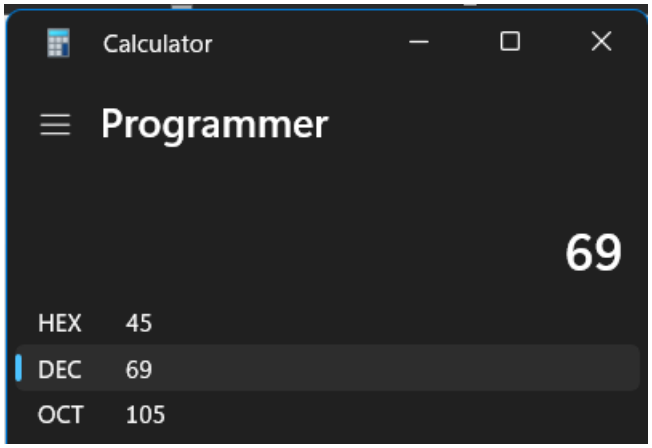
Register Output Data : R22:R21 = 01:05

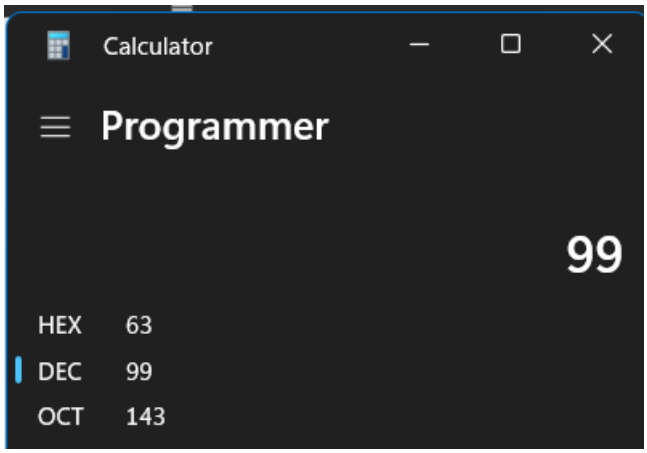


Register Input Data : R20 = 99

Register Output Data : R22:R21 = 01:43

- **Pembuktian**





- **Kode Program (Metode Lookup Table)**

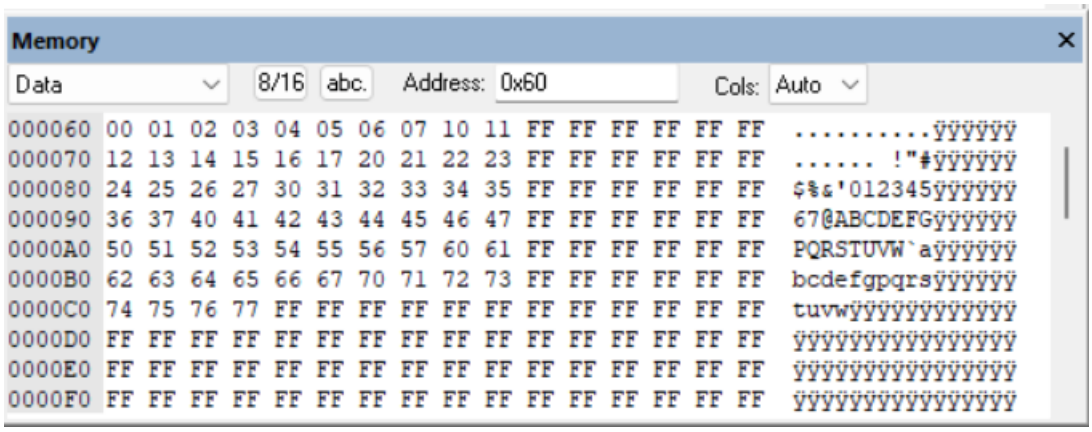
```
.include "m8535def.inc"
.org 0x00
rjmp main

main :
ldi r16,low(ramend)
out spl,r16
ldi r17,high(ramend)
out sph,r17

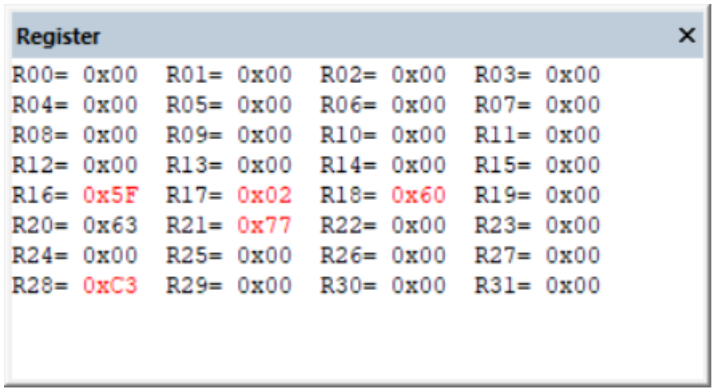
ldi r18,0X60
mov r28,r20
add r28,r18
adc r29,r30
ld r21,y

akhir : rjmp akhir
```

- **Output Program**



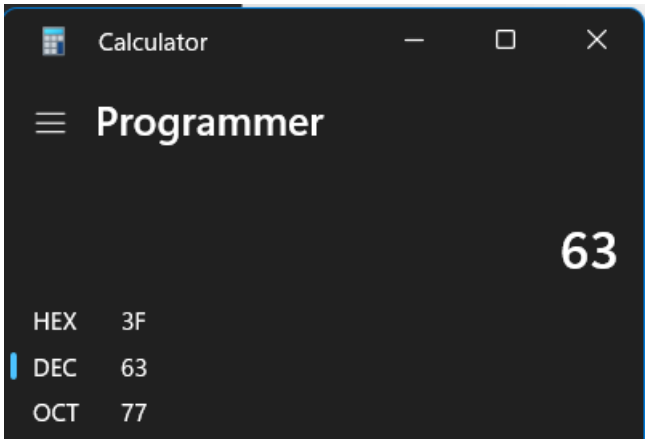
Data Area yang Berisi Nilai Konversi Desimal Menjadi Oktal



Register Input Data : R20 = 63

Register Output Data : R21 = 77

- **Pembuktian**



2. Buat program Konversi Data Oktal 2 digit (yang ada di R20) ke Desimal. Simpan hasil konversi pada R21.

- **Kode Program**

```
.include "m8535def.inc"
.org 0x00
rjmp main
      ; PAKAI PROJECT "ROOT"
main :
ldi r20,low(ramend)
out spl,r20
ldi r20,high(ramend)
out sph,r20
```

```
ldi r20,0x44                ; data input
mov r22,r20
andi r22,0xf0               ; mengambil nilai delapanan
lsr r22
lsr r22
lsr r22
lsr r22
rcall kal                   ; penggandaan 3 kali
mov r21,r22

mov r22,r20
andi r22,0x0f

add r21,r22
mov r19,r21
rcall daa
mov r21,r19

akhir: rjmp akhir

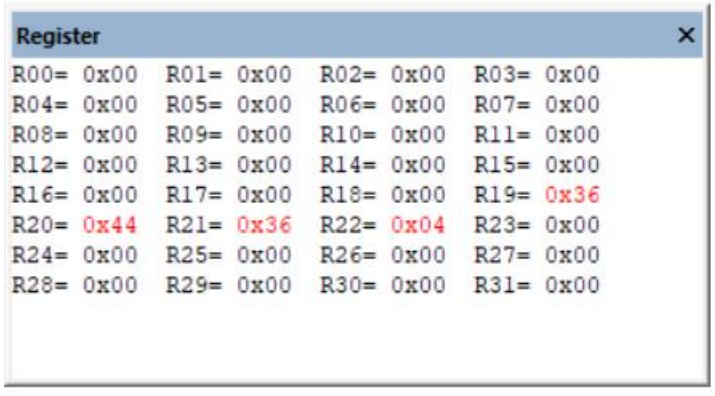
kal:        ; subrutin perkalian 8 secara desimal = penggandaan secara 3 kali
ldi r16,0x03
l1:   add r22,r22
      mov r19,r22
      rcall daa
      mov r22,r19
      dec r16
      brne l1
ret

daa:        push r16;amankan isi reg. ke stack
            push r17
            push r21
            push r24
```

```
            in r16,sreg; amankan sreg
            in r17,sreg
            ldi r24,0 ; r24 hanya sbg status dr carry flag
            adc r24,r24; status carry flag sebelum koreksi
; ambil, amankan kembali sreg, uji half carry
            out sreg,r17
            brhc lup4; lompat ke lup4 jika H=0
            rcall fk_ln; satuan + 06 jika H=1
      ;     rjmp lup5
lup4:       mov r21,r19
            andi r21,$0f; ambil dgt.satuan
            cpi r21,$0a ; > 9
            brcs lup5; lompat ke lup5 jika < 9
            rcall fk_ln; satuan + 06 jika > 9
lup5:       out sreg,r17; ambil, amankan kembali sreg, uji carry
            brcc lup6; lompat ke lup6 jika C=0
            rcall fk_hn; puluhan + 60 jika C=1
      ;     rjmp lup7
lup6:       mov r21,r19
            andi r21,0xf0; ambil dgt.puluhan
            cpi r21,0xa0
            brcs lup7; lompat ke lup7 jika pul < a
            rcall fk_hn; pul + 60 jika > 9
lup7:       out sreg,r16; mengatur carry flag sblm keluar:
            clc; C = 0
            in r16,sreg ;amankan sreg stlh c=0
            cpi r24,0 ;menguji status
            breq lup8
            out sreg,r16;ambil sreg, set c
            sec; C = 1
            in r16,sreg; amankan sreg stlh c=1
lup8:
            out sreg,r16;status akhir
            pop r24;kembalikan isi reg. dr stack
```

```
                pop r21
                pop r17
                pop r16
                ret
fk_ln:          ldi r21,06; koreksi dgt.satuan
                out sreg,r16
                add r19,r21
                in r16,sreg
                brcc tuh
                ldi r21,0x01
                add r17,r21
tuh:out sreg,r16
                adc r24,r24
                ret
fk_hn:          ldi r21,$60; koreksi dgt.puluhan
                out sreg,r16
                add r19,r21
                in r16,sreg
                adc r24,r24; amankan carry pd r24
                ret ; akhir dari prog. subrutin.
```

- **Output Program**



Register Input Data : R20 = 44

Register Output Data : R21 = 36

- **Pembuktian**

